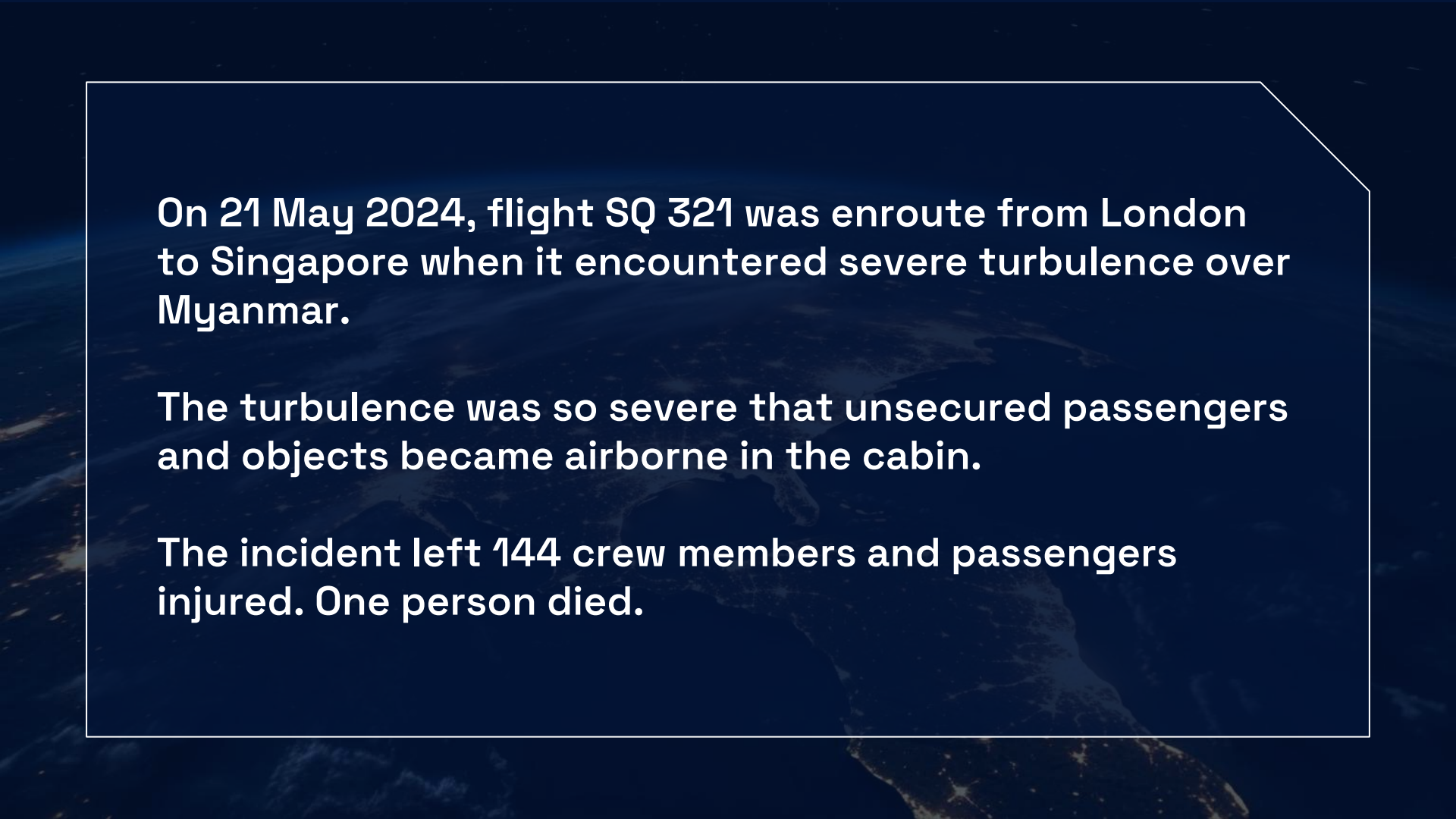


Taming The Skies:

Predicting Air Turbulence Using Machine Learning



Chee Qian Wen



On 21 May 2024, flight SQ 321 was enroute from London to Singapore when it encountered severe turbulence over Myanmar.

The turbulence was so severe that unsecured passengers and objects became airborne in the cabin.

The incident left 144 crew members and passengers injured. One person died.



**Singapore Airlines flight
hits severe turbulence**

A Singapore Airlines flight

Why was the consequence so severe?

Initial findings indicate that the “fasten seat belt” sign was turned on too late.

This is because current methods for detecting turbulence are largely reactive rather than predictive.

This means that turbulence is usually detected in real-time as it happens, rather than based on forecasts.

Overview



01

Introduction

- What is turbulence?
- Why should it be predicted?

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The Approach

- Data sources
- Method and success metrics

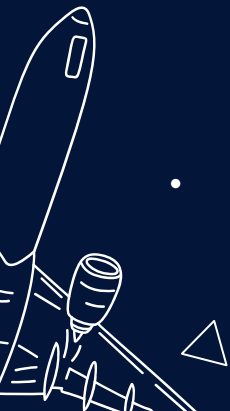
03

Analyses

- What influences turbulence?
- Trends and patterns

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Modelling

- Predicting turbulence using machine learning
- 

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Recommendations

- Actionability
- Future work

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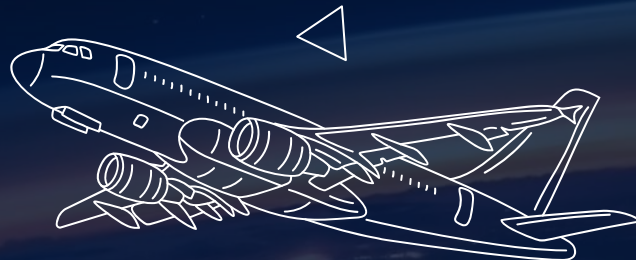
Conclusion

- Key takeaways
- Problem, solution, impact

01

Introduction

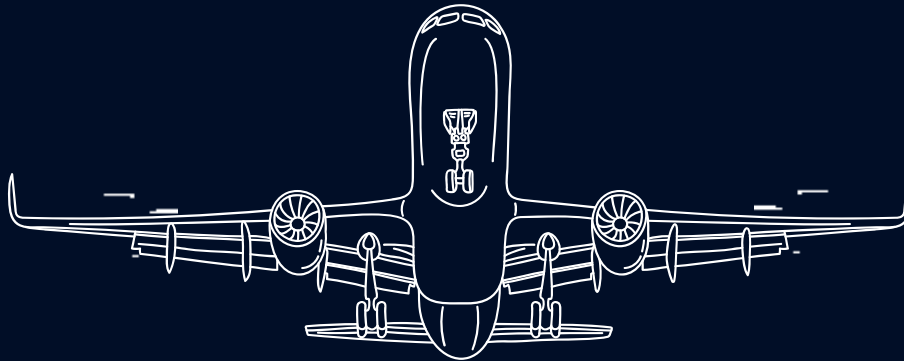
- What is turbulence?
- Why is it important to predict turbulence?





What is turbulence?

Turbulence occurs when the smooth airflow around an aircraft gets disrupted, causing sudden, unpredictable movements.



Why must turbulence be predicted?

Due to climate change, incidents of air turbulence are on the rise, causing significant concern amongst airlines and passengers.



Impact of Air Turbulence



Pilots

Greater responsibility →
Need to plan and adjust
flight paths



Passengers

Stressful experience →
Fear of flying and risk of
injuries



Airlines

Higher costs → Delays,
payouts, maintenance due
to wear and tear



Air Traffic Control

More crowded airspace →
Frequent re-routing of
flights, delays, and costs

According to the U.S. Federal Aviation Administration (FAA),
turbulence costs U.S. airlines approximately US\$100 million annually.

Solution

Examine the features that influence turbulence and build a model that can accurately predict the severity of turbulence based on these features





02



The Approach

- Data sources
- Method and success metrics

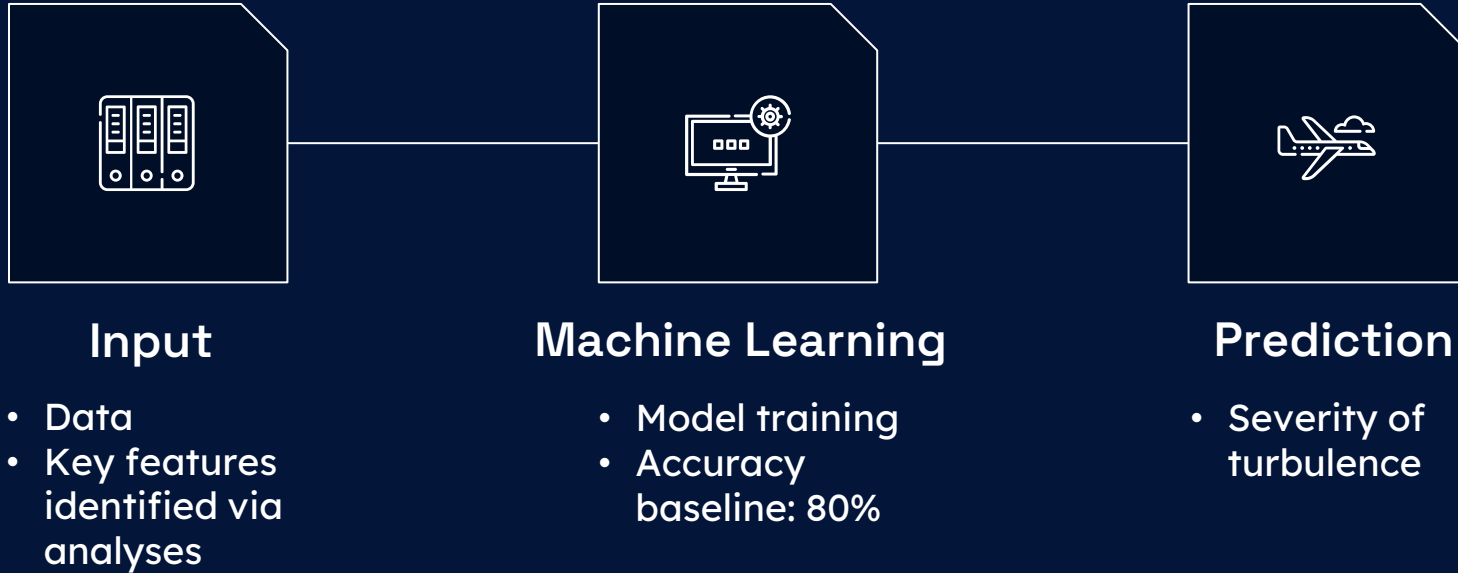
Data Sources

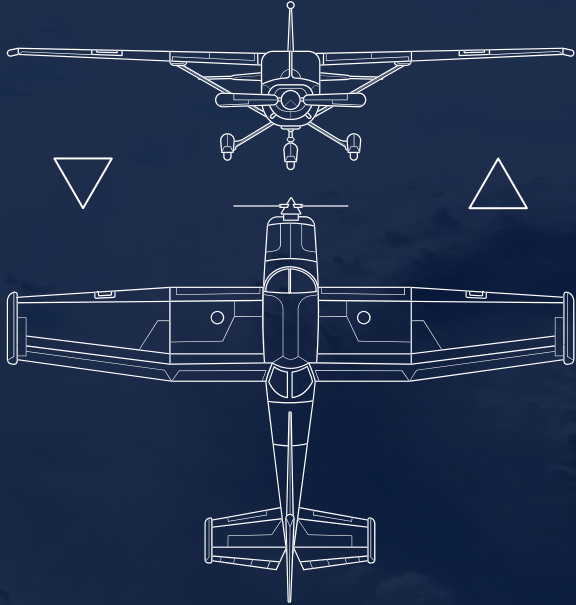
- This analysis focused on turbulence over U.S. mainland over the past 10 years (2015 – 2024)
- All data sourced from official U.S. government agencies

Dataset	Key Features	Data Source
Pilot Reports of Turbulence (PIREPs)	Date, time, location, aircraft type, severity	U.S. Federal Aviation Administration (FAA) (archived by Iowa State University Environmental Mesonet)
Aircraft Specifications	Model, weight class, engine count	International Civil Aviation Organization (ICAO)
Ground Elevation Data	Elevation	U.S. Geological Survey (USGS)
Metereological Data	Temperature, wind, humidity	U.S. National Weather Service (NWS) (archived by Iowa State University Environmental Mesonet)

- In total, over 1.5 million reports of turbulence were analysed

Method and Success Metrics





03

Analyses

- What influences turbulence?
- Trends and patterns



Number of Incidents

- Number of turbulence reports have been on the rise over the past 10 years



Number of Incidents

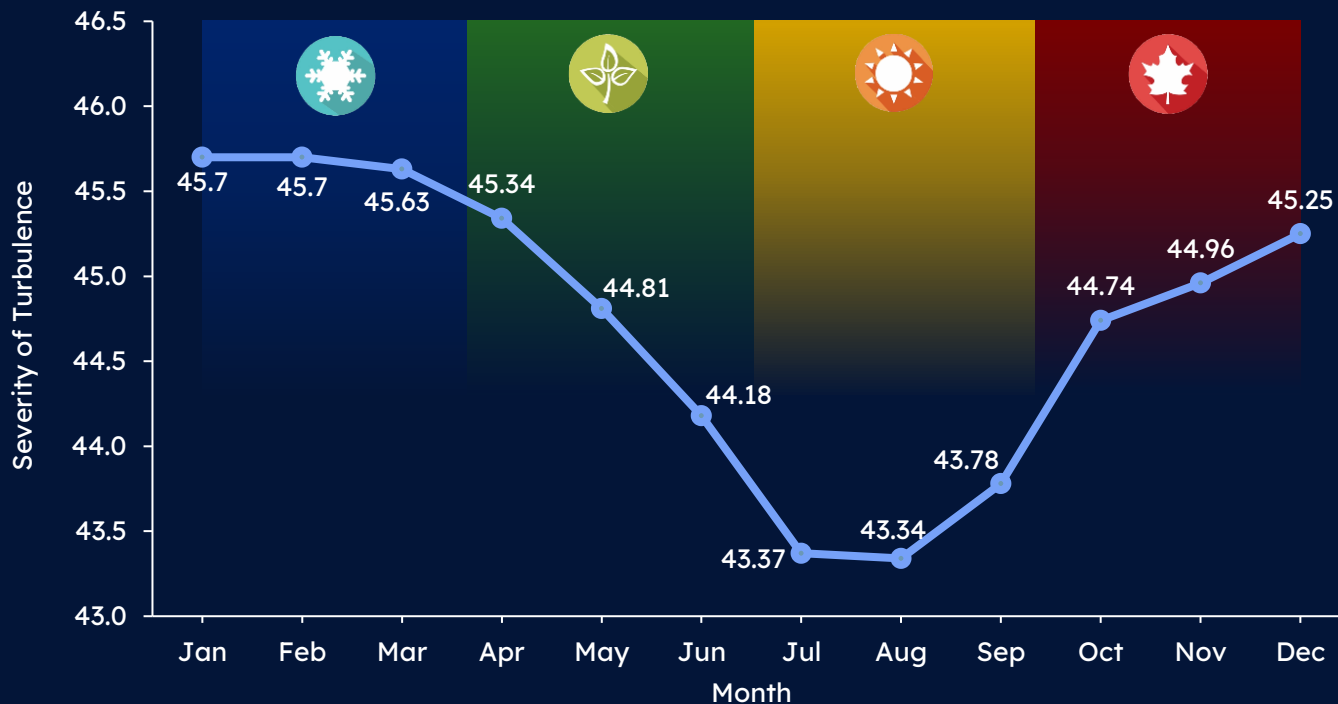
- Uptick driven mostly by an increase in light turbulence over time
- Occurrence of severe to extreme turbulence remains minimal





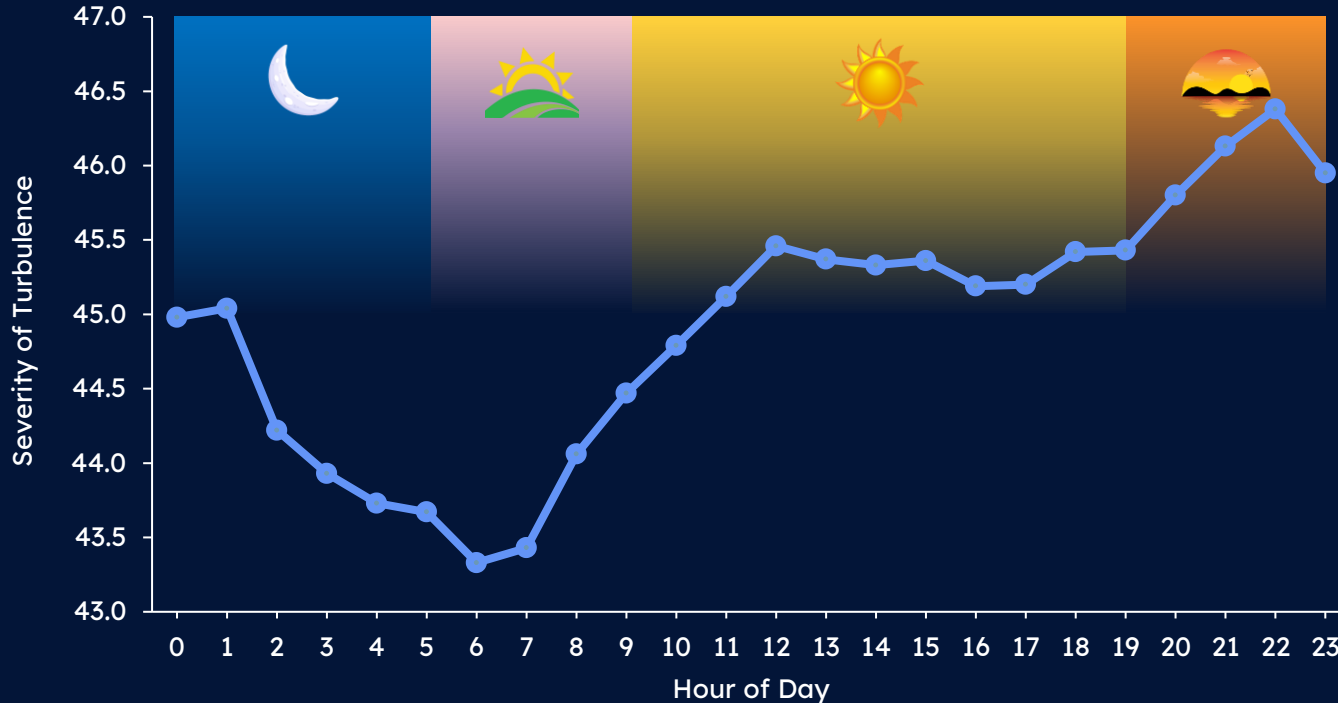
Seasonal Variations

- Turbulence is worst in winter due to stronger jet streams and powerful winter storms
- Turbulence is mildest in summer due to smaller storms and less wind shear



Daily Variations

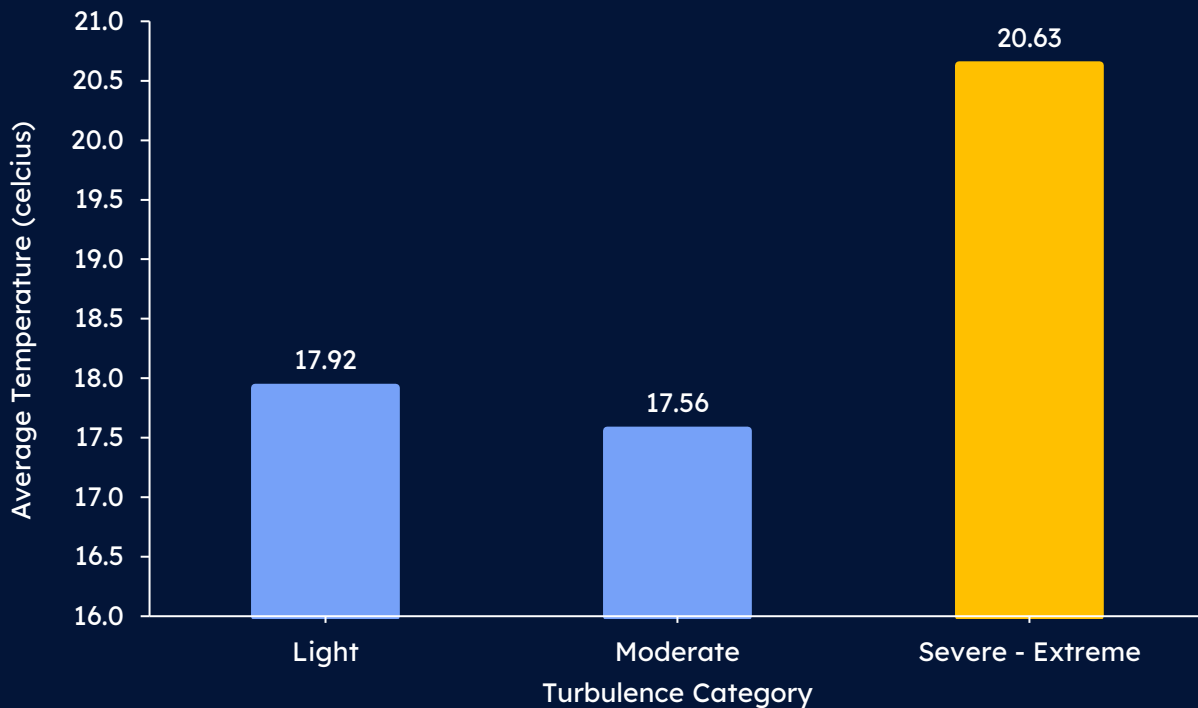
- Turbulence is mildest during the night as the Earth's surface cools
- Turbulence is worst during the day due to warm air rising up (updrafts)





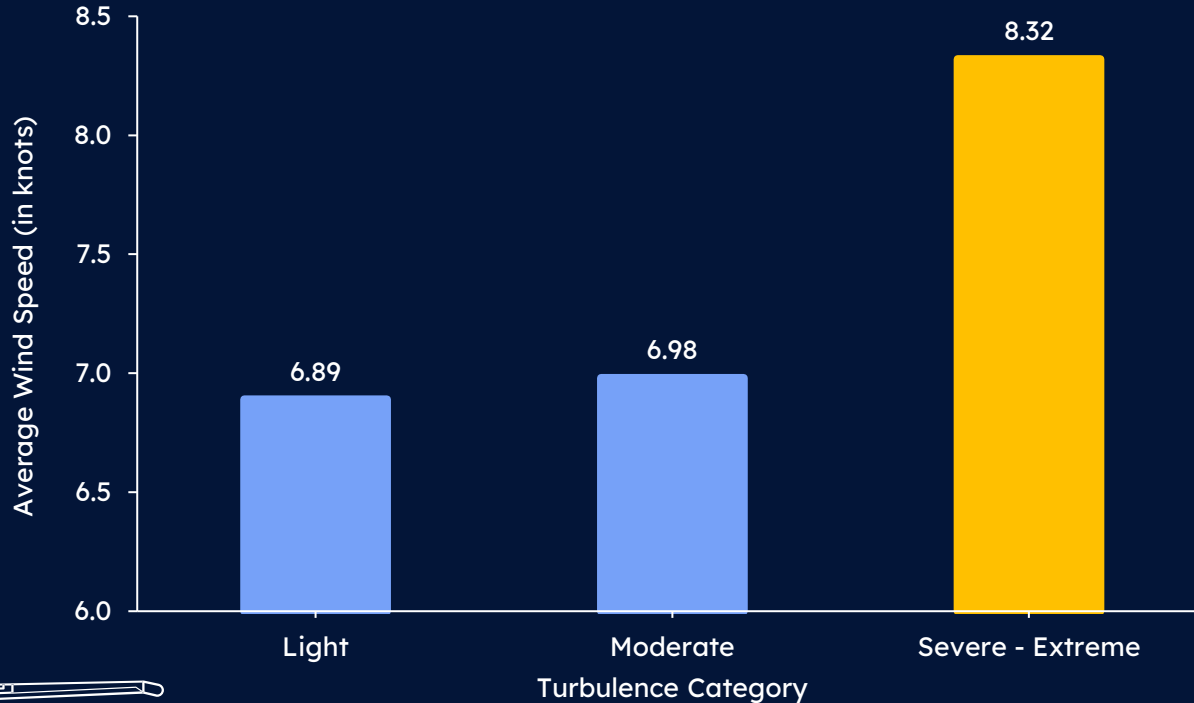
Temperature

- As temperature increases, severity of turbulence also increases
- Warm air is less dense and creates stronger updrafts, leading to more unstable air movement



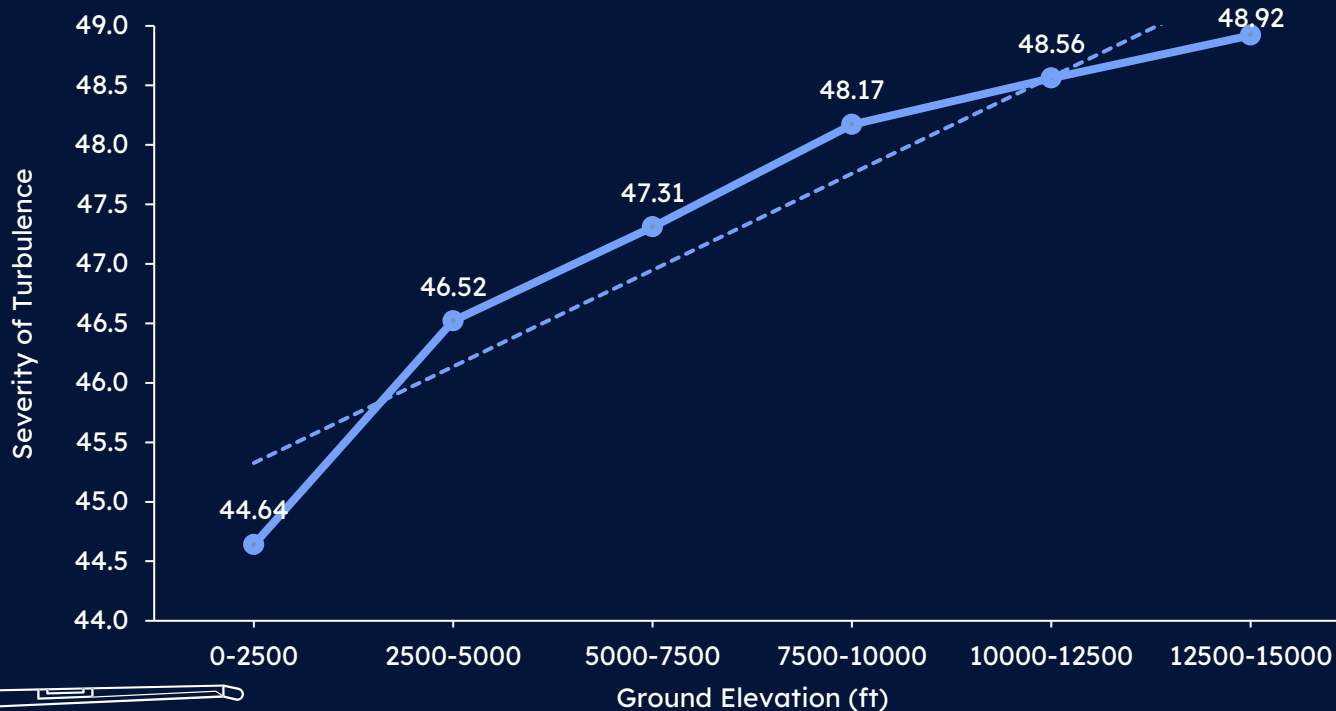
Wind Speed

- As wind speed increases, severity of turbulence also increases
- Faster winds create stronger air disturbances and wind shear, leading to unstable airflow

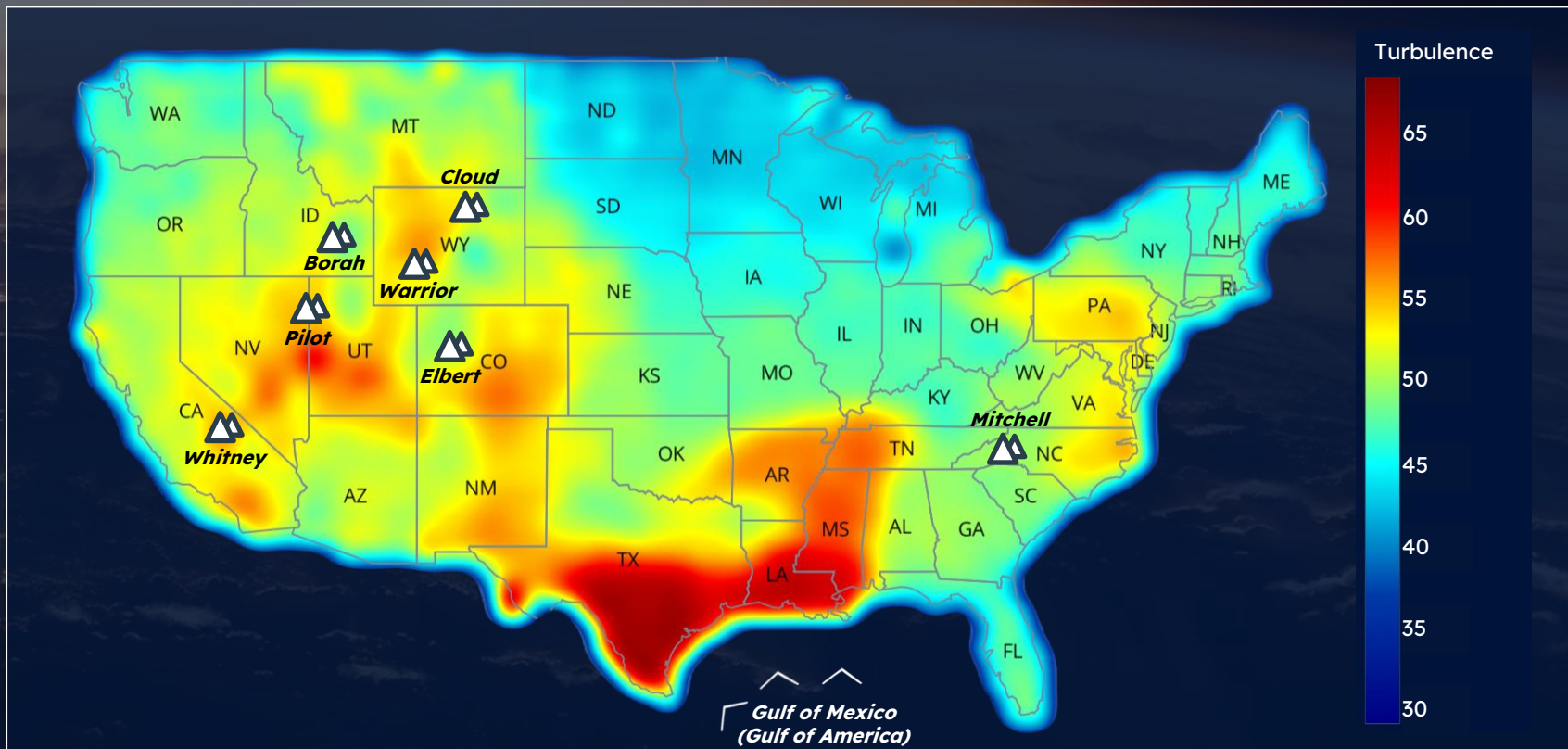


Ground Elevation

- Turbulence is worse when the ground is elevated and/or terrain is rough
- Wind speed changes rapidly due to friction between the air and the Earth's surface



Putting It All Together...





Why machine learning?

Turbulence is the result of a *complex interaction* between factors like ground elevation, wind speed, and temperature.

With machine learning, the computer can be trained to recognize patterns and make decisions to predict turbulence based on massive amounts of historical flight data.

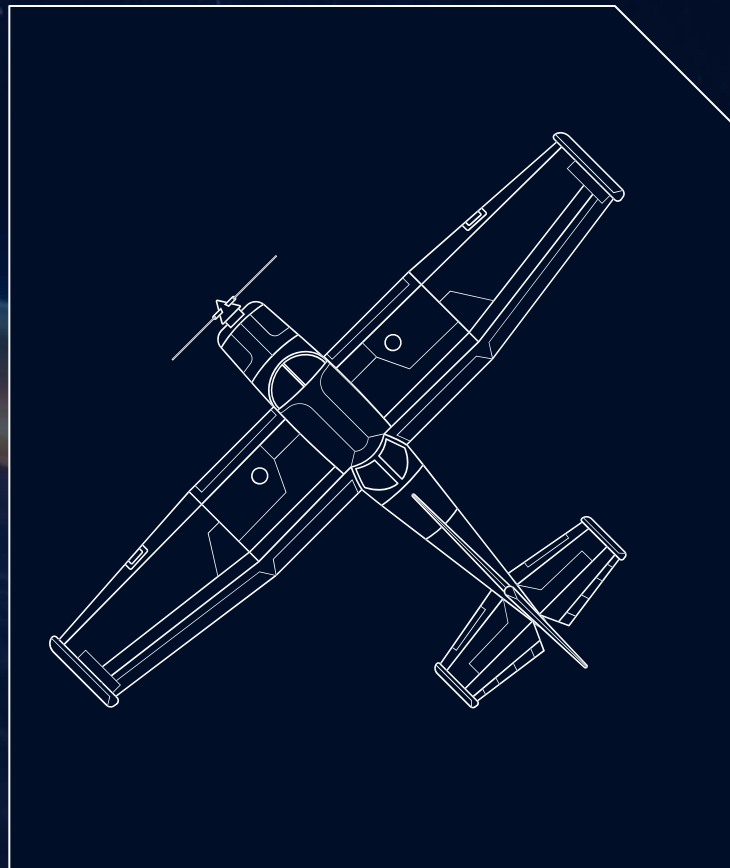
This allows it to identify conditions that typically lead to turbulence.



04

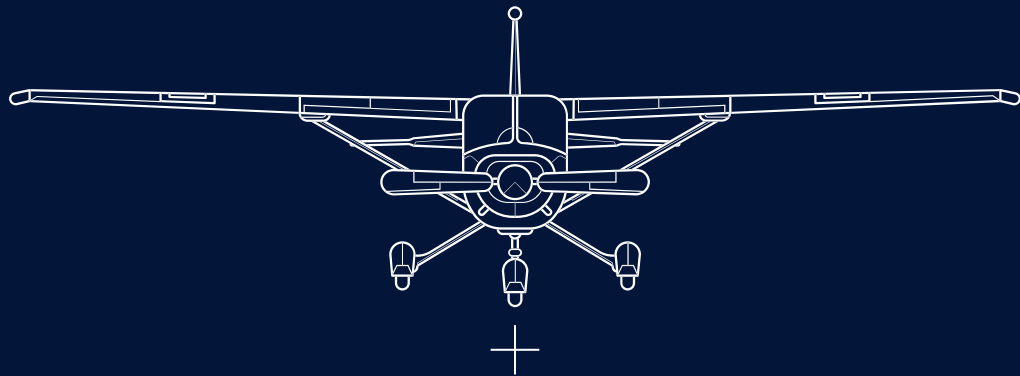
Modelling

- Predicting turbulence using machine learning



Modelling Process

- Aim of the models was to learn from the dataset
- Based on identified features, the models had to predict and classify the severity of turbulence into 3 categories: Light, Moderate, or Severe
- Models were trained on a training dataset and tested on unseen data
- Goal was to achieve at least 80% predictive accuracy

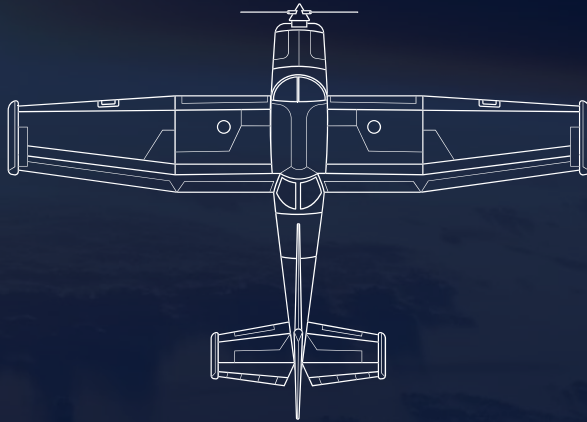




Model Comparisons

- 4 models were built and evaluated:

Model	Train Accuracy	Test Accuracy	Train-test Difference	Runtime
Decision Tree	99.66%	69.84%	29.82%	20.6s
XGBoost	81.85%	81.58%	0.27%	1m 28s
CatBoost	81.66%	81.57%	0.09%	3m 54s
Random Forest	81.67%	81.58%	0.09%	1m 50s



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Recommendations

- Actionability
- Future work

An Integrative Approach



- Aircrafts currently do not have onboard predictive turbulence systems
 - Current systems (e.g., aircraft sensors, radars) only detect turbulence in real-time



- There is also no single dedicated turbulence display on the pilot dashboard
 - Turbulence is inferred from separate systems for wind, temperature, etc.
 - Pilots must manually interpret turbulence risk from several systems



- To enhance safety and efficiency, it is recommended that airlines consider integrating AI-driven prediction tools into pilot dashboards to anticipate and forecast turbulence in advance



Future Work & Improvements



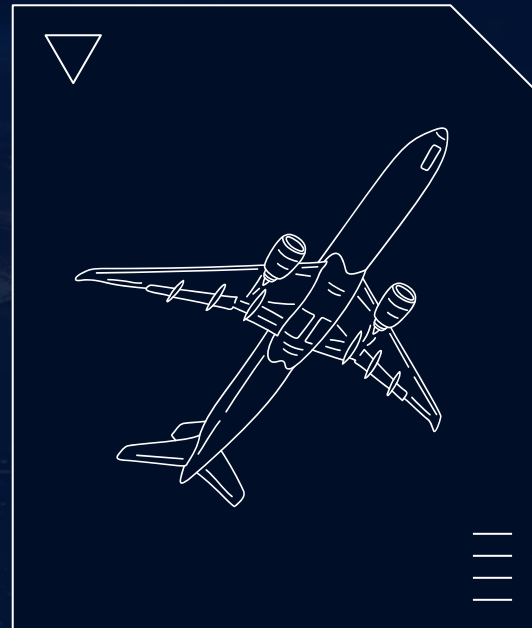
- To enhance model accuracy, future research should explore improving the variety of data used:

Wind	• Additional metrics, such as wind shear and rotational airflow, can provide further insights into turbulence • Available via aircraft sensors and satellite data
Satellite data	• Satellite data on weather conditions at different altitudes would be more reflective of flight conditions
Beyond U.S.	• Atmospheric conditions vary globally • Expanding the dataset to include oceanic, polar, and international airspace can enhance model accuracy

06

Conclusion

- Key takeaways
- Problem, solution, and impact

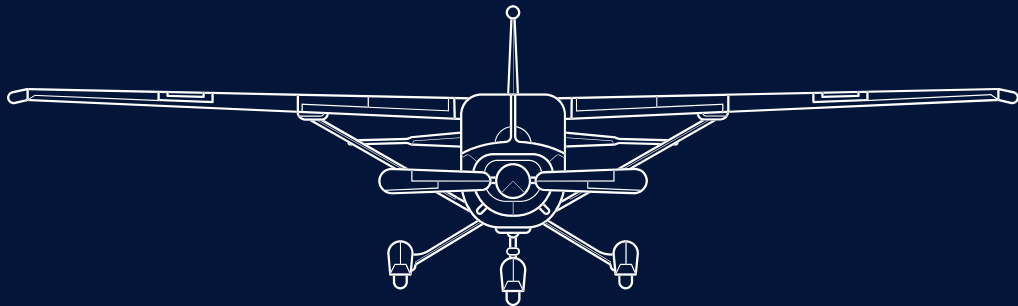




Key Takeaways



- This analysis examined a vast dataset containing over 1.5 million turbulence reports
- Several key features influencing turbulence were identified: Season, time of day, wind speed, temperature, and ground elevation
- By utilizing the XGBoost machine learning model, a relatively high predictive accuracy of 81.85% was achieved
- These findings serve as a proof of concept for the effectiveness of machine learning in predicting turbulence using meteorological data



Problem, Solution, and Impact



Problem

- Due to climate change, incidents of air turbulence are on the rise, causing significant concern to both airlines and passengers



Solution

- By utilizing machine learning models, turbulence can be predicted with relative accuracy by analyzing identified features, such as meteorological data

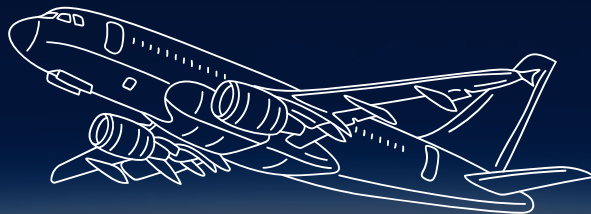


Impact

- Predicting and forecasting turbulence can help improve safety and efficiency, thus saving costs and minimizing risks to passengers



Thank You



Wishing everyone blue skies and fair winds!

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